

Router propagation: evidence and limits

Signal Space / GROSS Test 3

Technical execution: completed. Scientific classification: pass.

The explicit homogeneous router is compared with a derived Gram cone. Incidence and conversion scales are assumed. The complete SS OPS 1 spectrum and autonomous clocks are not measured.

Run: run-915ed88048939f8f

Analysis: analysis-0001-53e821db

Solver commit: 0babae84f204ac6445c5c11a0c8a0e3b0c771817

Config SHA-256: 7540ff4770e1e68b9255e38a7552f5e28c892ad68ecf4a1184e2e1bcadae1f24

Renderer commit: 0babae84f204ac6445c5c11a0c8a0e3b0c771817

Locked checks:

coverage: pass; value 624

fourier-circuit: pass; value 6.778545366739879e-16

conservation: pass; value 6.661343625597994e-16

continuum: pass; value 0.007742446435048791

triad-rank: pass; value [3, 3, 1]

order-control: pass; value 0.00987207275809493

rotation: pass; value 5.551115123125783e-16

group-refinement: pass; value 1.1180240583330558e-07

full-zone: pass; value 1.3322676295501878e-15

Method: exact displacement-stencil composition; independent matrix exponentials; analytic quaternion product and derivative; periodic real-space updates. All 26 signed axial/oblique directions at $|q|=0.02, 0.04, 0.08, 0.16$. Three triads, two orders, zone grids $16^3/32^3$ and random boxes $8^3/12^3$.

Error budget: floating-point roundoff, finite-difference group derivatives, finite sampling and contour interpolation. No numerical time integration. Conservation is wave amplitude norm; no physical detector record or mechanical energy-momentum inferred.

Next: Test 4 zero-wave complete tangent spectrum, retaining physical stationary memory variations. Only then attempt one bounded self-consistent nonzero periodic background. Strong shared-cone and geometric-wave-plus-material-memory hypotheses remain distinct.

cone-sections

Question: Which projector arrangements yield an isotropic, anisotropic or degenerate cone?

Reading: Gram ranks [3, 3, 1]; exact and leading contours are compared at phase 0.16. Rank check: pass.

Significance: Independent oblique projectors permit anisotropic propagation; collinear projectors remove two spatial principal directions although the internal matrix algebra is unchanged.

Limitation: A $q_z=0$ section, with interpolation, is not a full spatial surface or detector geometry.

The routing directions are assumed.

continuum-convergence

Question: Does the actual circuit approach its predicted cone as wavelength grows?

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Reading: Largest fine-radius residual 0.00774245; worst adjacent fine/coarse ratio 0.497147.

Continuum check: pass.

Significance: Shrinking finite-band error tests the derived Gram cone without fitting it to the measured dispersion.

Limitation: Four radii and 26 directions; this is a linear wave-sector limit, not a complete coupled-system continuum limit. Collinear roundoff values are not fitted.

order-and-direction

Question: Is opposite-direction asymmetry a finite-band order effect or leading drift?

Reading: Order-control status: pass; independent-triad minimum of maximum order differences 0.00987207 rad. Opposite-direction odd phase is divided by $|q|$ to expose any leading term.

Significance: The signed triple-sine term can generate directional asymmetry while the leading generator stays traceless. Order is physical incidence data.

Limitation: No leading-drift fit or invariant detector record is measured. Other backgrounds and modified routing laws remain open under Test 11.

full-zone-spectrum

Question: What additional low and pi-quasifrequency modes are hidden outside the origin?

Reading: The full-zone check is pass. Grid counts and exact corner/half-pi node probes are saved. The orthogonal circuit has zero and pi nodes away from the origin.

Significance: The near-origin Weyl cone cannot be interpreted as a unique physical species from the long-wavelength calculation alone.

Limitation: Finite grid inspection does not prove all nodes were found or measure topological charges. Negative wave bands are given by sign reversal; physical memory branches require Test 4.

group-directions

Question: Does the energy-transport direction follow the wavevector in the oblique case?

Reading: Derivative audit: pass. Maximum analytic-reference error 1.12e-07; step-halving difference 8.39e-08. Collinear nodal directions have no arrows.

Significance: Oblique projector geometry can direct group transport away from the wavevector direction without requiring several metrics.

Limitation: These are spectral group derivatives, not tracked wave packets. Arrows project onto xy and exclude touchings; full components remain in saved data.

numerical-audit

Question: Do explicit port shifts conserve amplitude and agree with the independent Fourier reference?

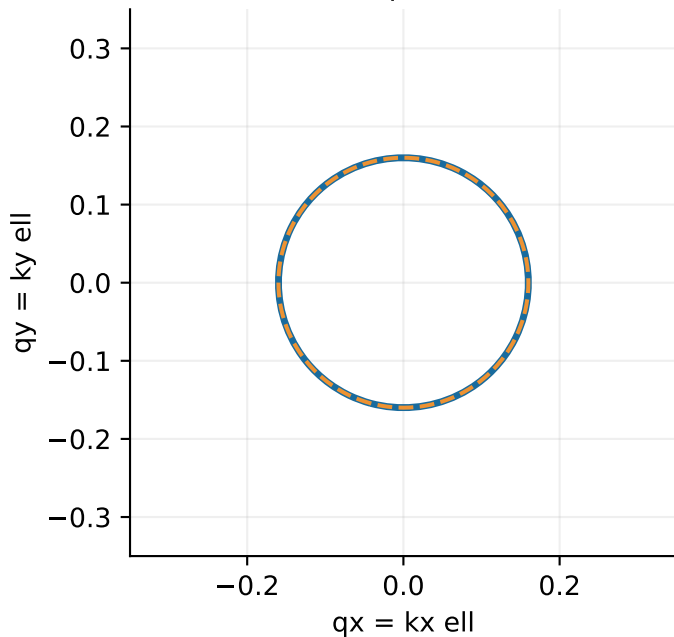
Reading: Fourier-circuit: pass; conservation: pass; common rotation: pass. Both box sizes and both block orders are included.

Significance: Independent real-space shifts and exponential/quaternion references test signs, order, amplitudes and global basis covariance.

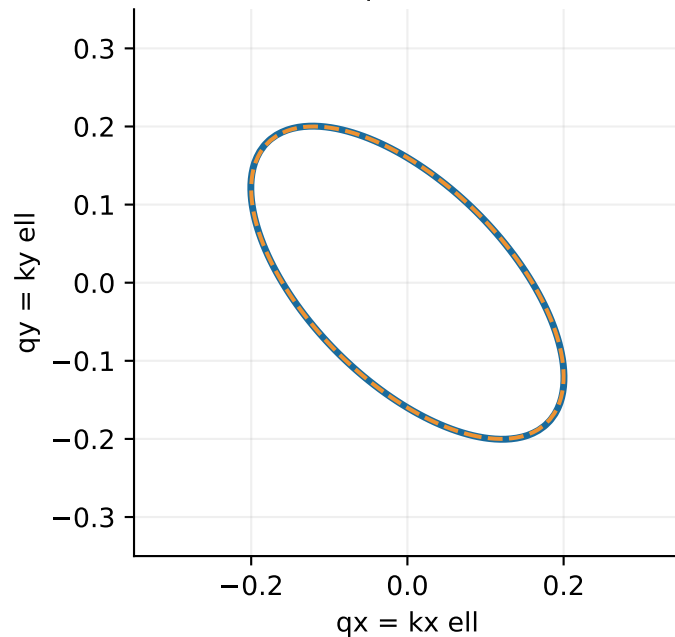
Limitation: Periodic boxes check implementation and norm only. There is no PDE timestep or outgoing boundary; no energy-momentum or nonlinear conservation claim follows.

Test 3 | Phase = 0.16 sections at $q_z = 0$
Blue: exact forward circuit; dashed orange: leading cone

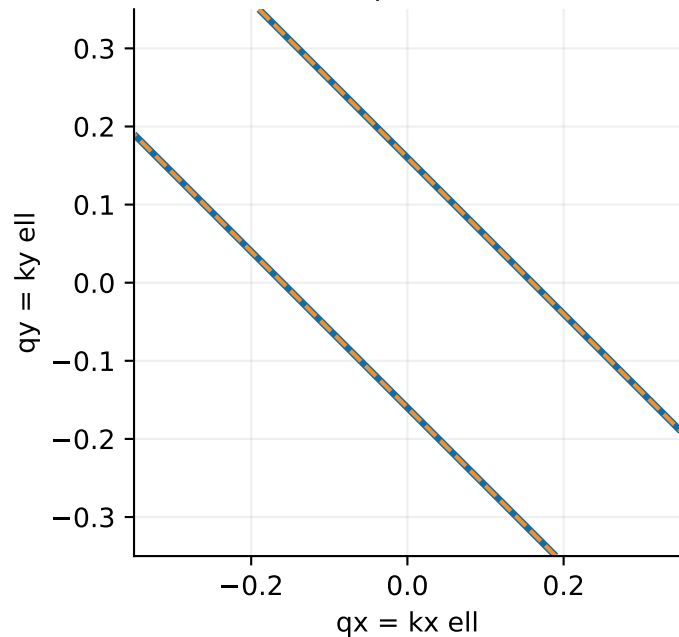
Orthogonal | Gram rank 3



Oblique | Gram rank 3

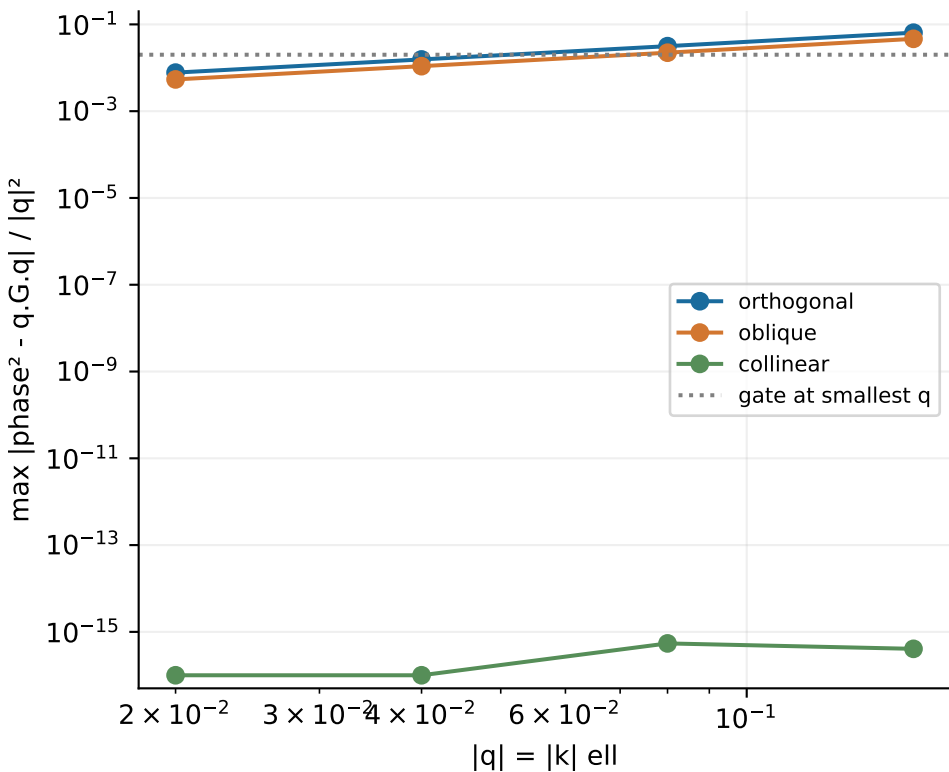


Collinear | Gram rank 1

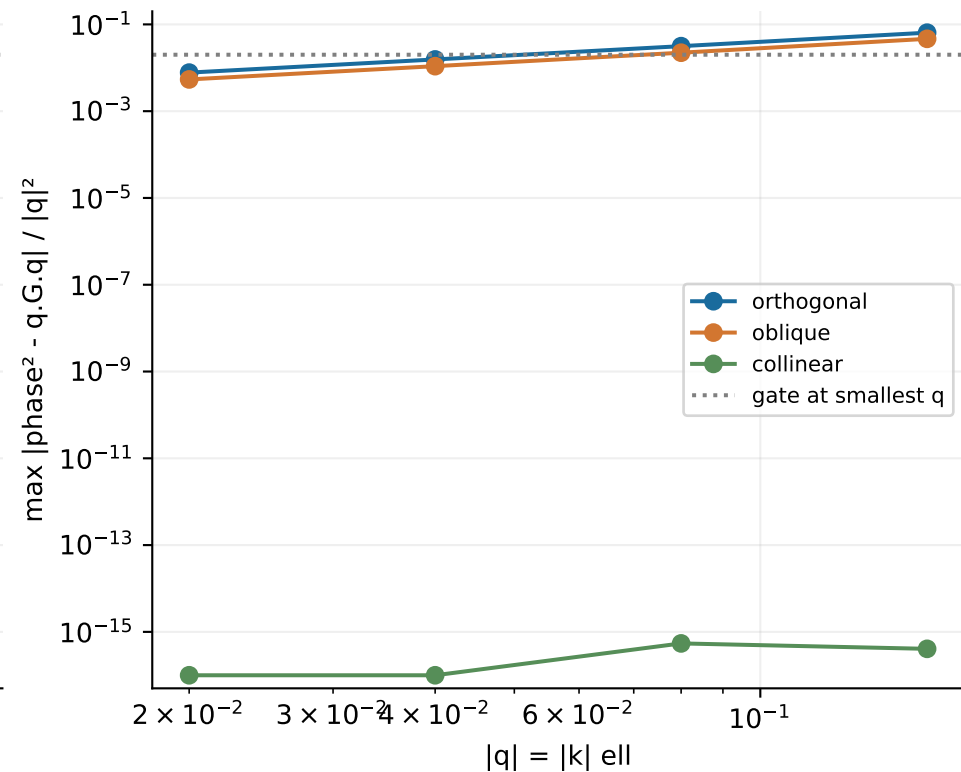


Test 3 | Leading-cone error across all 26 directions

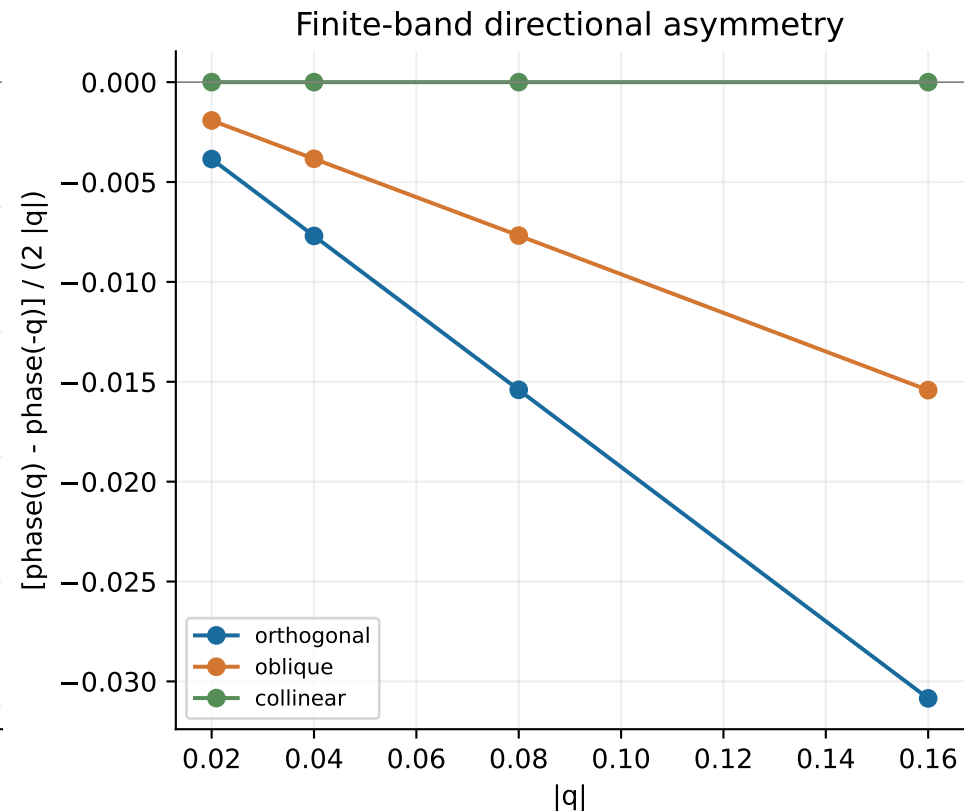
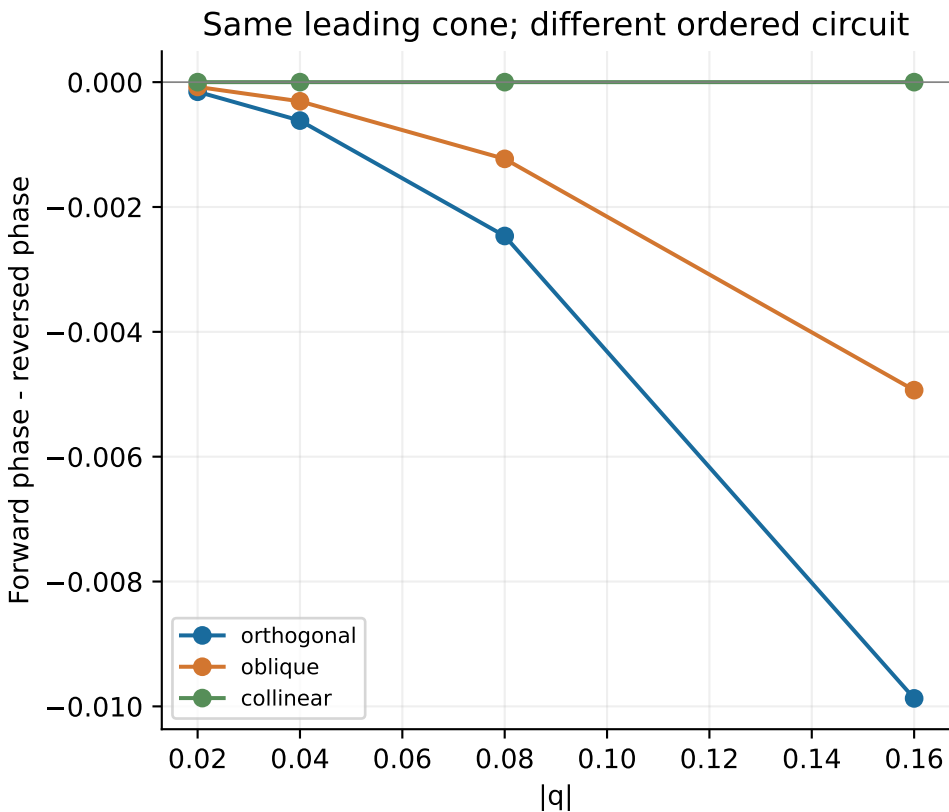
Forward order



Reverse order

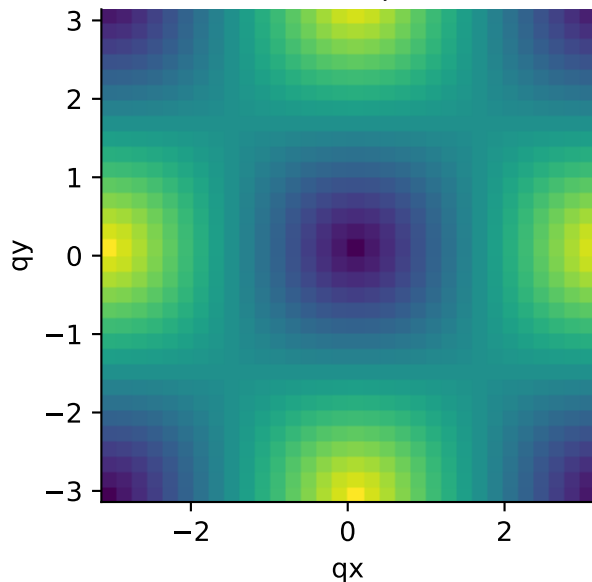


Test 3 | Fixed direction (1,1,1) / sqrt(3); no drift fit

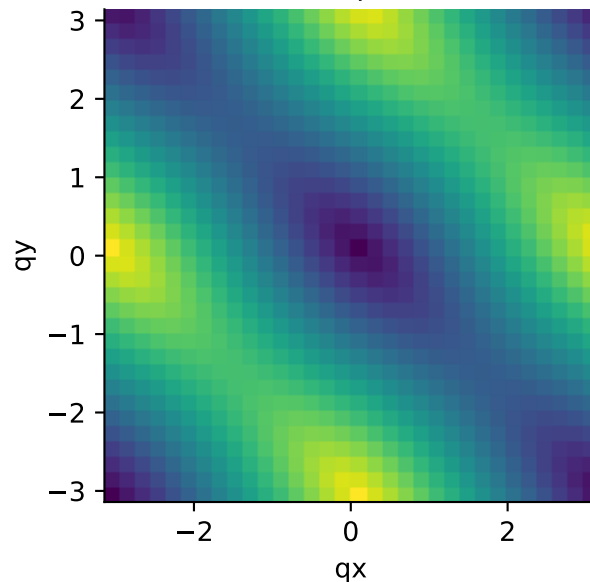


Test 3 | Full-zone wave bands: $\pm$ phase modulo 2π
Dark: zero quasiphase; bright: π quasiphase. Memories not included.

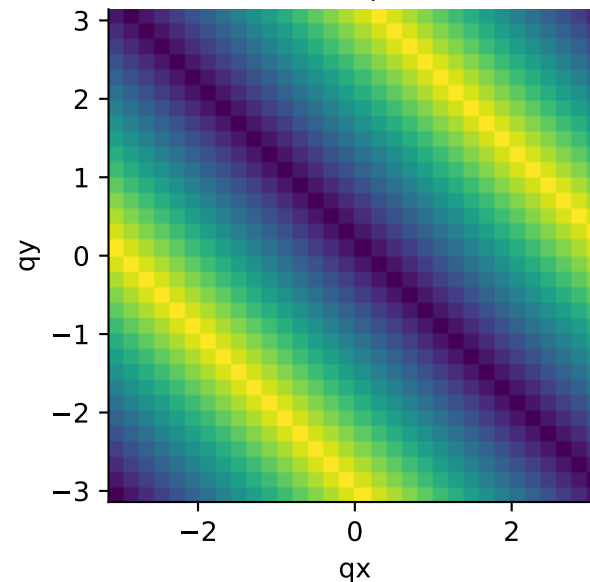
Orthogonal | $q_z = 0$



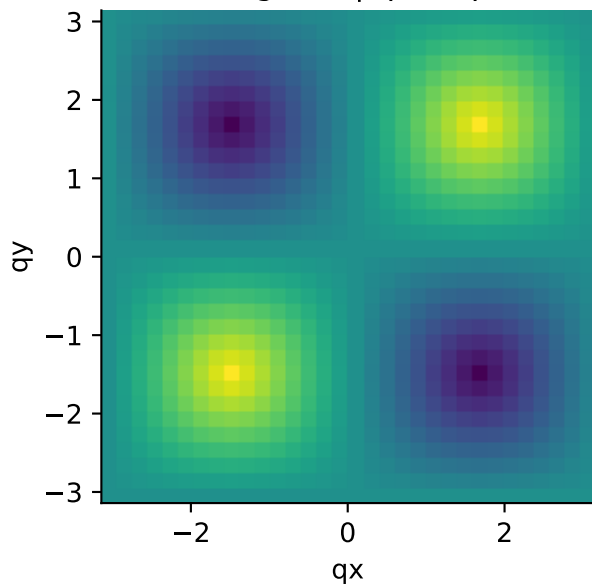
Oblique | $q_z = 0$



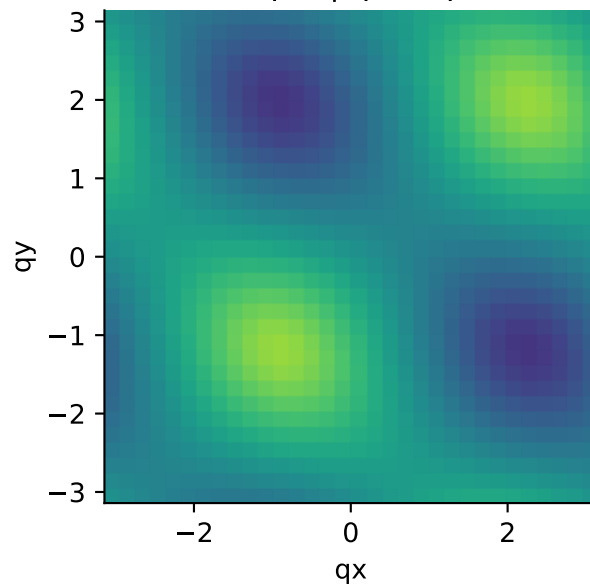
Collinear | $q_z = 0$



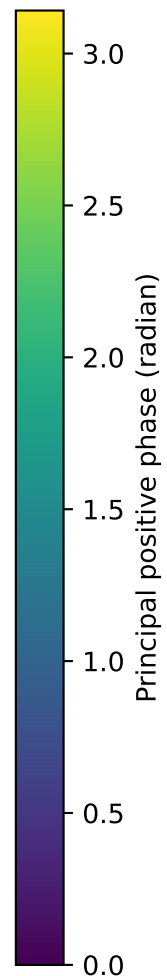
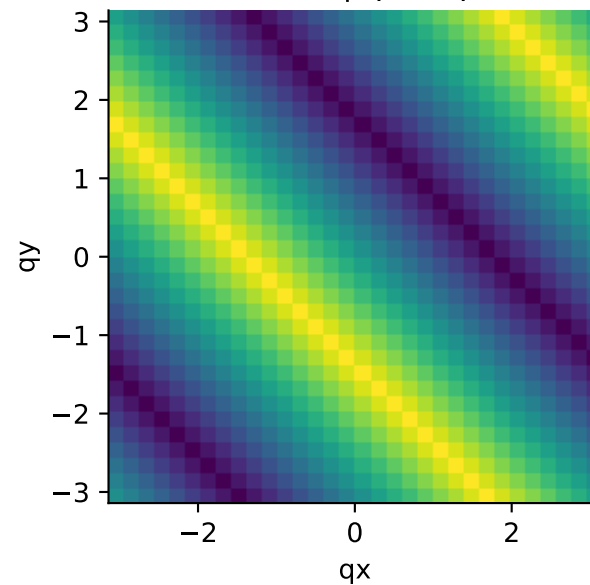
Orthogonal | $q_z = -\pi/2$



Oblique | $q_z = -\pi/2$

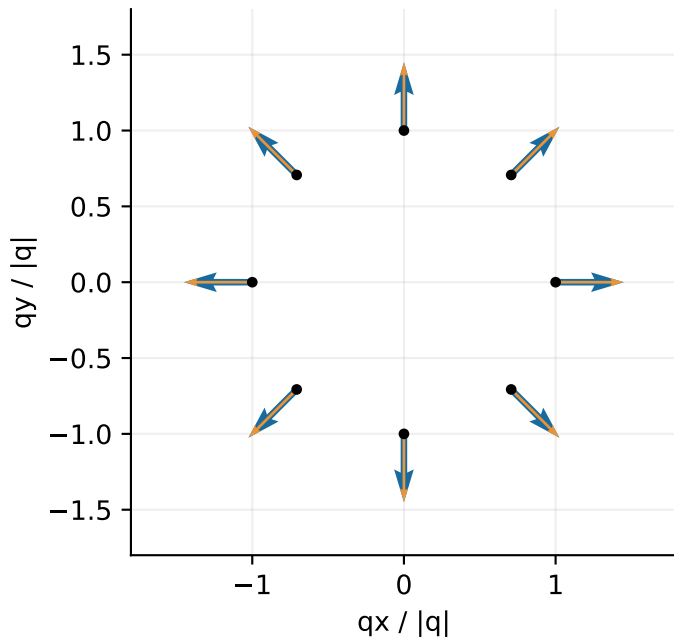


Collinear | $q_z = -\pi/2$

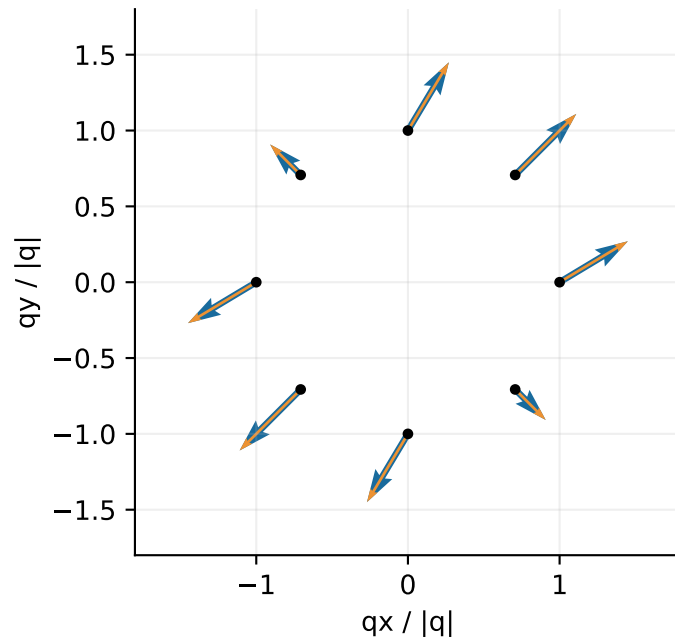


Test 3 | Group-velocity xy projections at $|q| = 0.02$
Blue: circuit derivative; thin orange: leading cone; vector scale 0.45

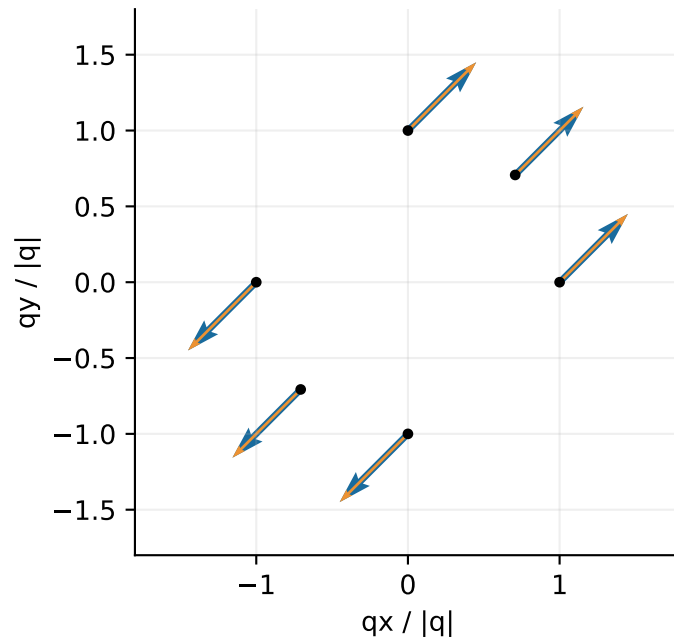
Orthogonal



Oblique

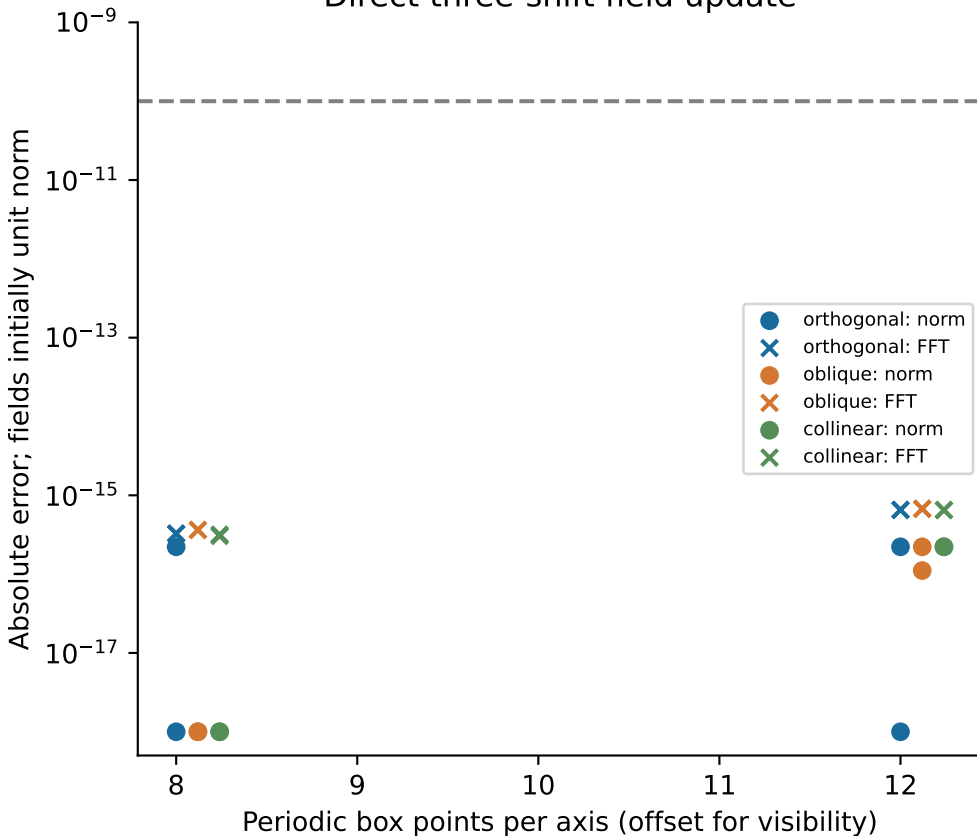


Collinear



Test 3 | Numerical accuracy controls; log floors retain raw zeros

Direct three-shift field update



Independent formula and derivative audits

